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SUBJECT CODE - DCA1105

SUBJECT NAME – FUNDAMENTALS OF MATHEMATICS

PROGRAM NAME - BACHELOR OF COMPUTER APPLICATIONS (BCA)

SEMESTER – 1

Q1.

(a)

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$$

Taking LCM,

$$= \lim_{x \rightarrow 0} \frac{(x - \sin x)}{x \sin x}$$

Now, using the standard limit:

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \frac{1}{6}$$

$$= \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \cdot \frac{x^2}{\sin x}$$

$$= \frac{1}{6} \times \lim_{x \rightarrow 0} \left(\frac{x^2}{\sin x} \right)$$

$$= \frac{1}{6} \times \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right) x$$

$$\text{Since } \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right) = 1$$

We get

$$= \frac{1}{6} \times 1 \times 0 = 0$$

Answer = 0

(b)

$$\lim_{x \rightarrow \frac{\pi}{2}} (\tan x - \sec x)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{\sin x}{\cos x} - \frac{1}{\cos x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - 1}{\cos x}$$

Multiplying numerator and denominator by $(\sin x + 1)$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(\sin x - 1)(\sin x + 1)}{\cos x(\sin x + 1)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin^2 x - 1}{\cos x(\sin x + 1)}$$

Using

$$\sin^2 x - 1 = -\cos^2 x$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\cos^2 x}{\cos x(\sin x + 1)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\cos x}{\sin x + 1}$$

Putting $x = \frac{\pi}{2}$,

$$= \frac{-0}{1+1} = 0$$

Answer = 0

Q2.

Rolle's Theorem states that if a function $f(x)$ satisfies the following three conditions on the interval $[a, b]$:

1. $f(x)$ is continuous on $[a, b]$.
2. $f(x)$ is differentiable on (a, b) .
3. $f(a) = f(b)$.

Then there exists at least one point c in (a, b) such that

$$f'(c)=0$$

In simple words, if a function starts and ends at the same value and is smooth without any breaks, then there must be at least one point between them where the tangent is horizontal.

Given Function

$$f(x) = \sin 2x$$

on the interval $\left[0, \frac{\pi}{2}\right]$

Step 1: checking continuity

The function $f(x) = \sin 2x$ is a trigonometric function and is continuous everywhere.

Therefore, it is continuous on

$$\left[0, \frac{\pi}{2}\right]$$

Step 2: Checking Differentiability

Since $\sin 2x$ is differentiable everywhere, it is differentiable on

$$\left(0, \frac{\pi}{2}\right)$$

Step 3: Verifying $f(a) = f(b)$

At $x = 0$,

$$f(0) = \sin(0) = 0$$

At $x = \frac{\pi}{2}$,

$$f\left(\frac{\pi}{2}\right) = \sin\left(2 \times \frac{\pi}{2}\right) = \sin \pi = 0$$

Thus,

$$f(0) = f\left(\frac{\pi}{2}\right) = 0$$

So the third condition is also satisfied.

Step 4: Find c such that $f'(c) = 0$

Differentiate the function:

$$f(x) = \sin 2x$$

$$f'(x) = 2 \cos 2x$$

According to Rolle's theorem,

$$f'(c) = 0$$

$$2 \cos 2c = 0$$

$$\cos 2c = 0$$

$$2c = \frac{\pi}{2}$$

$$c = \frac{\pi}{4}$$

Since

$$\frac{\pi}{4} \in \left(0, \frac{\pi}{2}\right)$$

the value lies within the given interval.

Q3.

(a) $\sin 75^\circ$

Since,

$$75^\circ = 45^\circ + 30^\circ$$

Using the formula:

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin 75^\circ = \sin(45^\circ + 30^\circ)$$

$$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

Substituting values:

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

Rationalizing:

$$= \frac{(\sqrt{3} + 1)\sqrt{2}}{4}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

Therefore,

$$\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

(b) $\cos 15^\circ$

Since,

$$15^\circ = 45^\circ - 30^\circ$$

Using the formula:

$$\begin{aligned}\cos(A - B) &= \cos A \cos B + \sin A \sin B \\ \cos 15^\circ &= \cos(45^\circ - 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ\end{aligned}$$

Substituting values:

$$\begin{aligned}&= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}}\end{aligned}$$

Rationalizing:

$$\begin{aligned}&= \frac{(\sqrt{3} + 1)\sqrt{2}}{4} \\ &= \frac{\sqrt{6} + \sqrt{2}}{4}\end{aligned}$$

Therefore,

$$\cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$$

Q4.

We know that the degree of the numerator is 3 which is greater than the degree of the denominator which is 2, therefore, we first perform **polynomial division**.

Step 1: Simplifying the Denominator

$$(x - 1)(x + 2) = x^2 + x - 2$$

So,

$$\frac{x^3}{x^2 + x - 2}$$

Step 2: Performing Division

Dividing x^3 by $x^2 + x - 2$:

$$x^3 \div x^2 = x$$

Multiply:

$$x(x^2 + x - 2) = x^3 + x^2 - 2x$$

Subtract:

$$x^3 - (x^3 + x^2 - 2x) = -x^2 + 2x$$

Now,

$$-x^2 \div x^2 = -1$$

Multiply:

$$-1(x^2 + x - 2) = -x^2 - x + 2$$

Subtract:

$$(-x^2 + 2x) - (-x^2 - x + 2) = 3x - 2$$

Therefore,

$$\frac{x^3}{x^2 + x - 2} = x - 1 + \frac{3x - 2}{(x - 1)(x + 2)}$$

Step 3: Partial Fraction Decomposition

Let

$$\frac{3x - 2}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}$$

Multiplying both sides by $(x - 1)(x + 2)$:

$$\begin{aligned} 3x - 2 &= A(x + 2) + B(x - 1) \\ 3x - 2 &= (A + B)x + (2A - B) \end{aligned}$$

Comparing coefficients:

$$\begin{aligned} A + B &= 3 \\ 2A - B &= -2 \end{aligned}$$

Adding both equations:

$$\begin{aligned} 3A &= 1 \\ A &= \frac{1}{3} \end{aligned}$$

Substituting into $A + B = 3$:

$$\begin{aligned} \frac{1}{3} + B &= 3 \\ B &= \frac{8}{3} \end{aligned}$$

$$\text{Answer} = \frac{x^3}{(x-1)(x+2)} = x - 1 + \frac{1}{3(x-1)} + \frac{8}{3(x+2)}$$

Q5.

Assuming the function is

$$y = a x^{nx}$$

where a and n are constants.

Using logarithmic differentiation:

$$\begin{aligned} \ln y &= \ln a + \ln (x^{nx}) \\ \ln y &= \ln a + nx \ln x \end{aligned}$$

Differentiating both sides with respect to x :

$$\frac{1}{y} \frac{dy}{dx} = n(\ln x + 1)$$

Therefore,

$$\frac{dy}{dx} = y n(\ln x + 1)$$

Substituting $y = ax^{nx}$:

$$\frac{dy}{dx} = ax^{nx} n(\ln x + 1)$$

$$\text{Answer} = \frac{dy}{dx} = ax^{nx} n(\ln x + 1)$$

Q6.

Let the length of the garden be x meters and the breadth be y meters.

Step 1: Using the Perimeter Condition

Given perimeter = 40 meters

$$2(x + y) = 40$$

$$x + y = 20$$

$$y = 20 - x$$

Step 2: Forming the Area Function

Area of rectangle:

$$A = x \times y$$

Substituting $y = 20 - x$:

$$A = x(20 - x)$$

$$A = 20x - x^2$$

Step 3: Differentiating the Area Function

$$\frac{dA}{dx} = 20 - 2x$$

For maximum area:

$$\begin{aligned}\frac{dA}{dx} &= 0 \\ 20 - 2x &= 0 \\ 2x &= 20 \\ x &= 10\end{aligned}$$

Step 4: Find the Breadth

$$\begin{aligned}y &= 20 - x \\ y &= 20 - 10 \\ y &= 10\end{aligned}$$

Step 5: Verify Maximum Area

Second derivative:

$$\frac{d^2A}{dx^2} = -2$$

Since

$$\frac{d^2A}{dx^2} < 0$$

the area is maximum.

Maximum Area

$$A = 10 \times 10 = 100 \text{ m}^2$$

Answer:

The area of the garden is maximum when:

Length = 10 m

Breadth = 10 m

Thus, the rectangular garden should be a **square of side 10 meters**, and the **maximum area is 100 square meters**.